

INTERPOLATION AND SMOOTHING OF BINOMIAL DATA FOR THE SOUTHERN AFRICAN BIRD ATLAS PROJECT

Lindsay McNeill: Department of Statistical Sciences, University of Cape Town, Rondebosch, 7700, South Africa.

Key words: Binomial; bird atlas; geostatistics; kriging; mapping; smoothing; spatial data.

Summary: In the Southern African Bird Atlas Project, reporting rates are mapped for quarter degree grid squares. For many squares the reporting rates are based on very few observations and are thus subject to considerable sampling error; for a number of squares there are no observations at all. The geostatistical technique known as kriging, commonly used for the interpolation of continuous spatial data, is adapted to exploit the binomial structure of the reporting rates in order to obtain smoothed maps which take into account the widely varying sample sizes, and to provide estimates for the unsampled squares.

1. THE SOUTHERN AFRICAN BIRD ATLAS PROJECT

The Southern African Bird Atlas Project (Harrison, 1987; Underhill et al., 1991) has as its major objective to obtain and publish definitive information on the spatial distribution and relative abundance of individual bird species based on detailed field records. The spatial unit for data collection and mapping is the quarter degree grid 'square' (QDGS) of 15 minutes of a degree latitude by 15 minutes of a degree longitude, or approximately 28×24 km in the centre of South Africa.

The field data for the atlas are collected by observers, most of whom are volunteers, and the data are checked and collated by the atlas co-ordinator, with the help of a number of regional assistants. The observers complete monthly field cards for individual QDGS, on which the presence or absence

of each individual species is recorded. That is, if the species was sighted at least once during the month, it is recorded as present. The 'reporting rate', that is, the ratio of the number of cards indicating the presence of a given species to the total number of cards completed, is calculated for each QDGS and each month. While such ratios are not directly comparable *between* species, because some species are more readily observed and identified, they do provide information on the relative spatial abundance of any given species (Underhill, et al., in press).

Data for the atlas were collected over the five-year period 1987 to 1991; in this paper we make use of all the data that were available in April 1990 for the 1974 QDGS in South Africa, Lesotho and Swaziland. For some squares the total number of field cards completed and checked up to this time is quite large, while in more remote areas it is very small and, in some cases, zero (table 1). The project co-ordinators have endeavoured to obtain more field cards for sparsely sampled squares, but it is clear that there will always be squares for which there is relatively little available information.

Table 1. Frequency distribution of field cards as at April 1990

No. of cards	No. of QDGS
0	139
1 - 5	592
6 - 10	277
11 - 20	273
21 - 50	339
51 and over	354
	1974

In preparing the detailed map for each species to be published in the atlas, it is intended to use the aggregated data for the entire period; that is, to map the overall reporting rate for each QDGS, calculated as the ratio of the *total* number of monthly field cards showing the species as present to the *total* number of field cards completed for that QDGS; the small number of

cards available in some areas does not allow for more detailed maps for individual months, although histograms showing the average reporting rate for each month, taken over all squares, will be included in the published atlas to give an indication of the seasonal variation of abundance.

For the purpose of this paper, we have chosen to look at the data for the Pied Crow *Corvus albus*, a striking and well-known bird with a fairly widespread distribution in southern Africa, except in the central Kalahari basin and in south-east Namibia (Maclean, 1985). A map of the raw reporting rates (figure 1) shows clear evidence of spatial continuity in the data and suggests four major areas of concentration of the Pied Crow in South Africa.

In view of the small number of field cards completed for many of the QDGS the raw reporting rates are subject to considerable sampling variation and it is thus appropriate to smooth the data values prior to final mapping.

2. KRIGING

The technique of *kriging* is commonly used to obtain a best linear unbiased predictor of a spatially correlated random variable. The name *kriging* was popularized by Matheron (1963) but the technique was developed independently by Gandin (1963). In most applications kriging has been used to interpolate the data values, i.e. the data are assumed to be error free. In practice this assumption is often unreasonable and in such cases one would not expect the estimator to reproduce the data values exactly, but rather to provide a smoothed version of the original data. When the variance of the measurement error is known, the kriging equations are readily adjusted to provide the optimum *measurement error-free* predictor. Cressie (1988) discussed the case when the variance of the measurement error is constant.

For binomial data, the measurement error is dependent on both the true probability and the sample size. It can, however, be estimated from the data.

In the data set considered here both the observed reporting rates and the sample sizes, that is, the number of completed field cards, vary very considerably between the QDGS and thus the difference in measurement error cannot be ignored, nor can it be removed by data transformation. Thus the standard kriging methods do not apply. In addition, the empirical variogram of the raw data, which is generally used to obtain an estimate of the spatial correlation function, is not a good estimator in the case of binomial data.

We deduce appropriate kriging equations and suggest a modified variogram estimator which is appropriate for binomial data.

3. THE MODEL

We assume that, for a given bird species, the reporting rate in the i -th QDGS would tend to some value π_i as the number of completed field cards tends to infinity. Thus π_i can be loosely interpreted as the probability that a 'typical' observer in the i -th QDGS will see the given species in any month. In practice such a probability could vary between observers and also from month to month but we ignore this in the current model and consider π_i simply as a measure of the abundance of the given species in the i -th QDGS. The vector of values $\underline{\pi} = (\pi_1, \pi_2, \dots, \pi_{1974})$ can be modelled as a multivariate random variable with a spatial distribution, that is, we assume that the correlation of two values will depend in some way on their relative spatial locations. In practice one would expect the correlation to be a decreasing function of distance.

Thus for the i -th QDGS we define:

$$(1) \quad R_i = X_i / n_i = \pi_i + \varepsilon_i$$

where n_i is the number of completed field cards for the i -th square and X_i is the number of these on which the given species is recorded as present; thus R_i is the observed reporting rate. The distribution of X_i , conditional on π_i , is Binomial (n_i, π_i), and we assume that, if $i \neq j$, then X_i and X_j are conditionally independent, given π_i and π_j .

A simple model for the spatial structure of the π_i is obtained by assuming second-order stationarity, that is:

$$(2) \quad E[\pi_i] = \mu$$

$$(3) \quad \text{var}[\pi_i] = \sigma^2 = \rho(0)$$

$$(4) \quad \text{cov}(\pi_i, \pi_j) = \rho(d_{ij})$$

where $\rho(d_{ij})$ is some function of the distance d_{ij} between the i -th and j -th squares.

We also have the restriction that $\pi_i \in [0, 1]$.

The area of the QDGS decreases gradually as one moves to the south, from 28×25 km to 28×23 km, and in practice one might expect the variance to be larger for smaller areas; however, the differences in areas are relatively small and have been ignored in this analysis.

We have made a number of simplifying assumptions in the model as described above; these are discussed in more detail in section 7.

4. THE KRIGING EQUATIONS

We seek a best unbiased linear estimator of each of the π_i , that is, for each i , we seek an estimator of the form $\sum_{j=1}^N w_j R_j$, where the summation is over all N of the QDGS for which data are available, and the w_j are chosen to minimize

$$E \left[\left[\sum_{j=1}^N w_j R_j - \pi_i \right]^2 \right]$$

subject to the constraint $\sum_{j=1}^N w_j = 1$, the latter condition guaranteeing unbiasedness. Using the Lagrange multiplier technique it is easy to show that, for each i , the solution is given by the following equation (Journel & Huijbregts, 1978):

$$(5) \quad \begin{bmatrix} \frac{w}{-\lambda} \end{bmatrix} = \begin{bmatrix} K & \frac{1}{0} \\ \frac{1}{1} & 0 \end{bmatrix}^{-1} \begin{bmatrix} c \\ 1 \end{bmatrix}$$

where $k_{jk} = \text{cov}(R_j, R_k)$ and $c_j = \text{cov}(R_j, \pi_i)$ and λ is a Lagrangian multiplier.

We show in the appendix that

$$(6) \quad \text{cov}(R_j, R_k) = \begin{cases} \rho(d_{jk}) & j \neq k \\ 1/n_j [\mu(1-\mu) + \sigma^2(n_j-1)] & j = k \end{cases}$$

and

$$(7) \quad \text{cov}(R_j, \pi_i) = \begin{cases} \rho(d_{ji}) & j \neq i \\ \rho(0) & j = i \end{cases}$$

Note that, since $\text{cov}(R_i, \pi_i) \neq \text{cov}(R_i, R_i)$, the vector $\underline{c}$ in equation (5) is not equal to the i -th row and column of the matrix K and thus the solution is *not* given by $w_i = 1$ so that we do *not* get exact interpolation at the data points, as is the case when the data are error-free.

5. THE SEMI-VARIOGRAM

The kriging equations are readily solved to provide the optimal estimator for each QDGS if the covariance function is known. In practice it is unknown and thus the function ρ needs to be estimated from the data. Geostatisticians prefer to model the *semi-variogram* rather than the covariance function itself, because it is not affected by the estimate of the mean and is less affected by trend in the data (Jowett, 1955).

The semi-variogram of the random variable π is defined as

$$\gamma_{\pi}(d) = \frac{1}{2} E[(\pi_{\underline{y}} - \pi_{\underline{y}+\underline{d}})^2]$$

where d is the length of the vector $\underline{d}$. It is easy to show that:

$$\gamma_{\pi}(d) = \rho(0) - \rho(d).$$

The semi-variogram is usually estimated by fitting a model to the empirical semi-variogram, that is, a graph of $\frac{1}{2}$ the squared difference of two data values plotted as a function of their separation distance. When there is measurement error in the data, the empirical semi-variogram has expectation equal to the semi-variogram of the error-free values plus the variance of the measurement error (Cressie, 1988). In this case, however, the measurement error variance, that is, the variance of ε_i in (1), is not constant across all QDGS, but depends on both n_i and π_i and thus the empirical semi-variogram is too noisy to provide a good estimator of γ_π .

We can show (see appendix) that

$$(8) \quad \frac{1}{2} E[(R_j - R_k)^2] = \rho(0) - \rho(d_{jk}) + \frac{1}{2} \sum_{r=j,k} \frac{\mu(1-\mu)}{n_r} - \frac{1}{2} \sum_{r=j,k} \frac{\sigma^2}{n_r}$$

and hence, in order to model the semi-variogram of the random variable π , we have plotted values of the *modified* semi-variogram.

$$(9) \quad \frac{1}{2} [(R_j - R_k)^2 - \sum_{r=j,k} \frac{\hat{\mu}(1-\hat{\mu})}{n_r} + \sum_{r=j,k} \frac{\hat{\sigma}^2}{n_r}]$$

where $\hat{\mu}$ and $\hat{\sigma}^2$ are estimates of the mean and variance of the π_i .

As a first estimate we have used:

$$(10) \quad \hat{\mu} = \frac{\sum X_i}{\sum n_i}$$

and

$$(11) \quad \hat{\sigma}^2 = \frac{\sum (R_i - \hat{\mu})^2 - \hat{\mu}(1-\hat{\mu})\sum(1/n_i)}{N - \sum(1/n_i)}$$

where the summations are over all the N QDGS with $n_i > 0$. The variance estimator is based on the fact that

$$(12) \quad E\left[\sum_{i=1}^N (R_i - \mu)^2\right] = \sum_{i=1}^N \text{var}(R_i) \\ = \mu(1-\mu) \sum_{i=1}^N (1/n_i) + \sigma^2 \left[N - \sum_{i=1}^N (1/n_i)\right].$$

The estimator of the mean is unbiased, but the estimator of the variance is not; the bias is a function of the variance of $\hat{\mu}$ which depends on the unknown $\rho(d_{ij})$. However, the fitted semi-variogram provides another estimate of σ^2 and if the difference between the two estimates is large the new value may be substituted in equation (9) and the process repeated until convergence is obtained. In this case, the original estimates of μ and σ^2 were 0,334 and 0,0716 respectively; the first iteration of the variogram fitting process gave σ^2 as 0,0712, and a second iteration gave σ^2 as 0,0711.

In plotting the empirical semi-variogram (figure 2) values are grouped into lag distance classes of 5 km and only the average of each class is shown on the graph. The distances between QDGS were calculated as the distances between their respective centres; since the centres of the QDGS are approximately on a square lattice the distances tend to cluster, thus the smallest observed distance is of the order of 26 km and the next group of distances is around $26 \times \sqrt{2}$ km. The 'waviness' of the graph at large lags is due to the existence of a small number of large areas of high density which can be seen in figure 1. In practice, it is the behaviour of the variogram for small lags which is more crucial to the kriging process and thus we have not been especially concerned with the fit of the model for large lags.

In order to ensure a positive-definite covariance function it is common practice to fit one of the standard models as described in Clark (1982) or Journel and Huijbregts (1978). In this application, the exponential variogram model, fitted by the weighted least squares method (Cressie, 1985), was found to provide a satisfactory fit. The exponential model is given by:

$$\gamma(d) = C(1 - \exp(-d/a))$$

where C is the 'sill', which also estimates the variance $\rho(0)$. The value of a in the final fitted semi-variogram (figure 2) was 80,19 km and the sill was 0,0711.

6. SOLVING THE KRIGING EQUATIONS

Substituting the fitted parameters into the kriging equations we obtain the w_i and hence the $\hat{\pi}_i$ for each of the 1974 QDGS. The computations were based on the method of Davis and Grivett (1984), which expresses the kriging estimator in terms of the inverse of the covariance matrix K , thus taking advantage of the positive-definite structure of this matrix. In theory, all the data values should be included in the estimator for each point; in practice, given N data points, this necessitates the inversion of an $N \times N$ matrix, so that when N is large it is more common to use 'local' kriging, i.e. only neighbouring data points are used in the estimation of each point. In this case $N = 1835$ (the number of squares for which data were available) so local kriging was used, and a 7×7 neighbourhood of squares was used as the local window to compute each estimated value. An advantage of local kriging is that, when there is trend in the data, local kriging yields similar estimates to 'universal' kriging, which does not make the assumption of stationarity of the data but rather models the trend separately from the residual stationary stochastic component; the two methods give different results only when one extrapolates beyond the range of the data (Journel & Rossi, 1989). A disadvantage of local kriging for this set of data is that the matrix K in (5) has to be recomputed and inverted separately for each point to be estimated; the fact that the diagonal elements of K involve the n_i prevents one from taking advantage of the fact that the data points are (approximately) on a square lattice so that the inter-point distances, and thus the off-diagonal covariances in K , remain roughly constant for each 7×7 window.

In the map of the kriged estimates (figure 3) the sharp discontinuities evident in the original map (figure 1) have been smoothed out. The degree of smoothing is relatively greater in the more remote areas of the country, where the number of field cards tends to be smaller.

The estimation process does not explicitly constrain the values to lie in the range $[0, 1]$. In practice, none of the estimates exceeded 1 and only 28 of

the 1974 estimates were less than zero, with the smallest being -0.004; these 28 values were simply adjusted to zero. An alternative approach which avoids the problem would be to use constrained optimization techniques. The possibility of working with transformed data was also explored. The logit transformation is a possible candidate. However, a large number of the reporting rates were zero and it was thus necessary to use a modified logit. Various adjustments were considered, for example, Cox's empirical logistic transformation (Cox, 1970) given by:

$$Z_i = \log \left[\frac{X_i + \frac{1}{2}}{n_i - X_i + \frac{1}{2}} \right]$$

However, the transformed values are then dependent on the n_i ; for example, if $X_i = 0$ then $Z_i = \log(\frac{1}{3})$ when $n_i = 1$ and Z_i approaches $-\infty$ as n_i gets large. This leads to instability in the variogram and the estimation process.

7. DISCUSSION

There are a number of possible criticisms of our model; we now discuss these and suggest some possible refinements and improvements.

The π_i may vary with time, due to seasonal effects caused by migration or long-term trends in species abundance. However, the five-year data collection period is considered sufficiently short for such long-term trends to have a relatively small effect. The number of field cards sent in for any QDGS does not vary much with time of year, so that the process of aggregating monthly data is not unreasonable, and serves the aim of providing a measure of relative spatial abundance.

It is also likely that the π_i will vary between observers, depending on their ability to identify the species in question, on their exact location in the QDGS, and on the time and effort they expend in locating species. This could lead to a distribution with variance greater than the usual binomial model. In addition, the fact that each observer may send in a card each

month, and that the cards are aggregated over months, means that there will often be more than one card from a given observer in the raw reporting rate, which may violate the assumption of independent observations.

We have assumed that the error terms ϵ_i are not spatially correlated. This may not be so; for example, if many observers spend time at a site such as a large dam which straddles the boundary of two QDGS, this could cause R_i to overestimate π_i in both QDGS for certain species.

No particular distribution for the π_i has been assumed. In particular, if the distribution is very skew, then the linear kriging estimator may not be optimal. However, such skewness may be largely a result of spatial trends in the data. It may therefore be more appropriate to use a method of kriging which explicitly models trend. A simple parametric trend model such as a linear or quadratic function of latitude and longitude is not likely to be appropriate for the distribution of the reporting rates; Cressie (1986) has suggested a non-parametric method of trend removal which is suitable for gridded data. In any case, the use of local kriging as opposed to global kriging does effectively account for trend, and the variation of the π_i around the *local* means may well be reasonably symmetrically distributed. With this in mind, it may have been more appropriate to use a local rather than a global estimate of μ and σ in equation (9).

As in most kriging applications, we have chosen to assume stationarity of the covariance function. For species with very specific habitat requirements this may not be entirely appropriate and it may be necessary to regionalize the data in order to obtain regions in which the assumption of stationarity is more realistic. Another possible way of obtaining improved estimates is to incorporate information on the habitat by using the method of co-kriging (Journel & Huijbregts, 1978); this assumes however that detailed habitat information is actually available. Yet another possibility is to incorporate other forms of prior information via Bayesian kriging (Omre, 1987).

In considering these various criticisms and suggestions one must bear in mind that the ultimate objective in this project is to provide maps for close

to 900 species and thus it is important that the resultant methodology does not make excessive demands of computer time for negligible gain in accuracy; thus further research will probably be directed largely at techniques for more efficient computation.

8. CONCLUSION

A variant of the method of kriging has been developed for the situation where the data are binomial in nature and contain measurement error, so that smoothing is required prior to final mapping. The measurement error in this data is considerable for the many QDGS having low sample sizes. The wide range of sample sizes leads to a corresponding wide range of error variances so that standard techniques which assume constant error variance are clearly not applicable.

The Pied Crow was chosen to illustrate the technique because it is fairly widespread in its distribution. For other species which are more localized or have very specific habitat requirements a more sophisticated model incorporating additional information may be required; it may also be necessary to model trends more explicitly for those species for which such trends are more marked.

APPENDIX

We assume that the distribution of X_i , conditional on π_i , is binomial (n_i , π_i), and that, if $i \neq j$, then X_i and X_j are conditionally independent, given π_i and π_j . The spatial structure of the π_i is assumed to be as given by equations (2) to (4).

We first establish equations (6) and (7):

$$E[X_i^k] = \int_0^1 E[X_i^k / \pi_i] f(\pi_i) d\pi_i$$

and thus, in particular,

$$E[X_i] = n_i \mu$$

$$E[X_i^2] = n_i \mu + n_i(n_i - 1)(\sigma^2 + \mu^2)$$

$$\text{var}(X_i) = n_i \mu(1 - \mu) + n_i(n_i - 1)\sigma^2$$

$$\text{var}(R_i) = \mu(1 - \mu)/n_i + \sigma^2(n_i - 1)/n_i$$

also,

$$\begin{aligned} E[X_i \pi_j] &= \int_0^1 \int_0^1 E[X_i \pi_j / \pi_i \pi_j] f(\pi_i \pi_j) d\pi_i d\pi_j \\ &= n_i [\rho(d_{ij}) + \mu^2] \end{aligned}$$

so that

$$\text{cov}(X_i \pi_j) = n_i \rho(d_{ij})$$

$$\text{cov}(R_i \pi_j) = \rho(d_{ij})$$

and, if $i \neq j$,

$$\begin{aligned} E[X_i X_j] &= \int_0^1 \int_0^1 E[X_i X_j / \pi_i \pi_j] f(\pi_i \pi_j) d\pi_i d\pi_j \\ &= n_i n_j [\rho(d_{ij}) + \mu^2] \end{aligned}$$

so that

$$\text{cov}(X_i X_j) = n_i n_j \rho(d_{ij})$$

$$\text{cov}(R_i R_j) = \rho(d_{ij})$$

Proof of equation (8):

$$\begin{aligned} E[(R_j - R_k)^2] &= [\text{var}(R_j) + \mu^2 + \text{var}(R_k) + \mu^2 - 2(\text{cov}(R_j R_k) + \mu^2)] \\ &= [\text{var}(R_j) + \text{var}(R_k) - 2\rho(d_{jk})] \\ &= \sum_{r=j,k} [\mu(1 - \mu)/n_r + \sigma^2(n_r - 1)/n_r] - 2\rho(d_{jk}) \\ &= 2[\sigma^2 - \rho(d_{jk})] + \frac{1}{2} \sum_{r=j,k} [\mu(1 - \mu) - \sigma^2]/n_r \end{aligned}$$

ACKNOWLEDGEMENTS

The author would like to thank Professor Les Underhill for initiating the project and Professor Walter Zucchini for many helpful discussions. James Harrison of the Southern African Bird Atlas Project provided the data, Neville Verlander assisted with programming and model checking, and Peter Martinez wrote the programmes to produce the maps. Thanks are also due to the anonymous referees for their helpful comments.

REFERENCES

- CLARK, I. (1982). *Practical geostatistics*. London: Elsevier.
- COX, D.R. (1970). *The analysis of binary data*. London: Chapman and Hall.
- CRESSIE, N. (1985). Fitting variogram models by weighted least squares. *Math. Geol.*, 17, 563-586.
- CRESSIE, N. (1986). Kriging nonstationary data. *J. Amer. Statist. Assoc.*, 81, 625-634.
- CRESSIE, N. (1988). Spatial prediction and ordinary kriging. *Math. Geol.*, 20, 405-421.
- DAVIS, M.W. & GRIVET, C. (1984). Kriging in a global neighbourhood. *Math. Geol.*, 16, 249-265.
- GANDIN, L.S. (1963). *Objective analysis of meteorological fields*. Leningrad: GIMIZ. (Translated from the Russian in 1965 by R. Hardin, Israel Program for Scientific Translations: Jerusalem.)
- HARRISON, J. (1987). The southern African bird atlas project. *S. Afr. J. Sci.*, 83, 400-401.
- JOWETT, G.H. (1955). Sampling properties of local statistics in stationary stochastic series. *Biometrika*, 42, 160-169.
- JOURNAL, A.G. & ROSSI, M.E. (1989). When do we need a trend model in kriging? *Math. Geol.*, 21, 715-739.

- JOURNEL, A.G. & HUIJBREGTS, C.J. (1978). *Mining geostatistics*. London: Academic Press.
- MACLEAN, G.L. (1985). *Roberts' birds of southern Africa*, 5th ed. Cape Town: John Voelcker Bird Book Fund.
- MATHERON, G. (1963). Principles of geostatistics. *Econ. Geol.*, 58, 1246-1266.
- OMRE, H. (1987). Bayesian kriging - merging observations and qualified guesses in kriging. *Math. Geol.*, 19, 25-39.
- UNDERHILL, L.G., OATLEY, T.B. & HARRISON, J.A. (1991). The role of large-scale data collection projects in the study of southern African birds. *Ostrich*, 62, 124-148.
- UNDERHILL, L.G., PRYS-JONES, R.P., HARRISON, J.A. & MARTINEZ, P. (in press). Seasonal patterns of occurrence of migrants in southern Africa using atlas data. *Ibis*.

Figure 1. Pied Crow - Raw Data.



Figure 1. Pied Crow - raw data

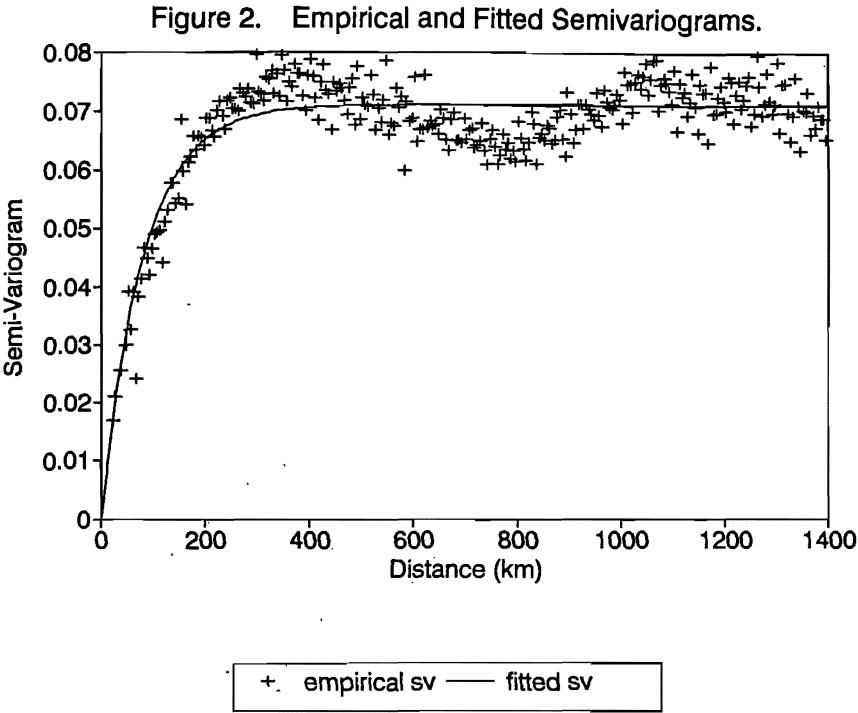


Figure 2. Empirical and fitted semivariograms

Figure 3. Pied Crow - Smoothed Data.

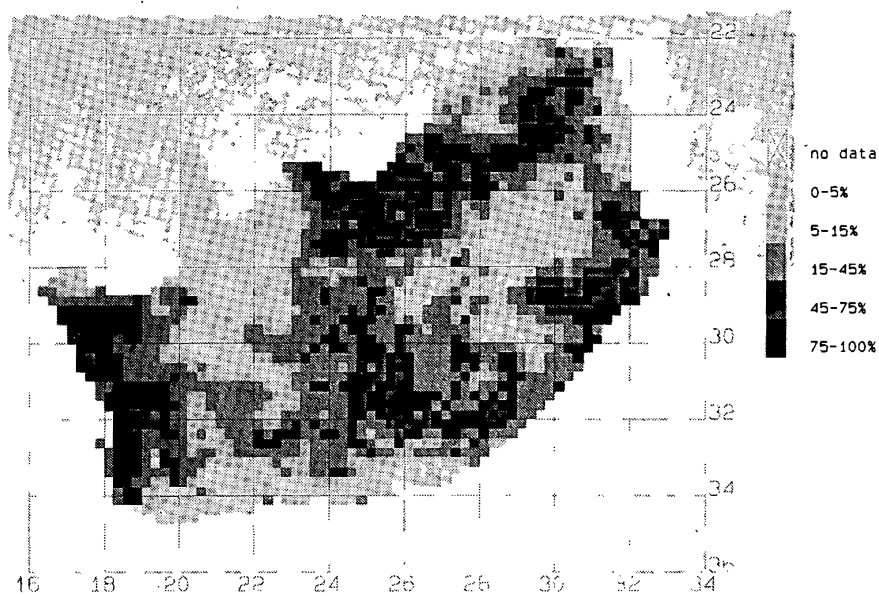


Figure 3. Pied Crow - smoothed data

Manuscript received, 21 January 1991, revised, 15 December 1991.